

building blocks of any 3D model irrespective of how complex. All shapes like rectangles, cubes and even curved surfaces are broken down into triangles. Here’s where the software comes in. Developers will use either of the computer graphics library software (OpenGL or Direct3D) to feed each triangle into the graphics pipeline, but one vertex at a time. The GPU can assemble vertices into triangles as and when needed.

These days due to the infinite Shader programming possibilities available, developers are able to give advanced light-based properties to objects. For example, the way water in a pond interacts with light, the surface sub-scattering of light by translucent materials of varying densities, like skin and hair, and more.

Stage 2: Transformations

GPUs have their own localised coordinate system and they can use this to specify the position of each object in a 3D scene. For utilising their own coordinate system the GPU has to convert all the objects to a common tracking system. At this point the GPU cannot perform complex modifications to the objects for fear of warping the triangles, simple operations like scaling and rotation can be done. By representing each vertex in a *homogenous* coordinate the GPU can perform a single operation on triangles *en masse*. At this stage of the pipeline, the output is a stream of billions of triangles on the basis of the GPUs common coordinate system in which the orientation is the viewers’ perspective.

Stage 3: Lighting

By now, each triangle is placed in a global coordinate system and the GPU can now calculate each triangle’s colour on the basis of the lights in the scene. The lighting could be as simple as a single source (a light bulb in a room), or as complex as a street lit up by a row of streetlights. In the latter case, the GPU is tasked with summing up the effects of each light source and computing their overall effect on colours and luminosity of the triangles in the scene. Today’s GPUs are also complex enough to colour surfaces of the basis of their physical surface properties—matte, gloss, etc. Advanced lighting techniques like specular highlighting, and diffusion and occluded lighting are also possible.

Stage 4: Getting Perspective

The next stage of the pipeline projects the (now) coloured triangles on to a virtual plane, as viewed from the users perspective. These coloured triangles and their respective coordinates are finally ready to be turned into pixels.

Stage 5: Rasterization

Rasterization is quite simply the process of converting a vertex representation to a pixel representation. Here, an image that is described as a vector graphics format is converted into a raster image consisting of dots or pixels. In this stage, each pixel can be treated separately, and the GPU handles all the pixels in parallel—one of the reasons that manufacturers upped the Shader unit count on GPUs to speed up this parallel processing.

Stage 6: Texturing

While pixels are already coloured by this stage sometimes additional textures may need to be added for added realism. This is a cosmetic process similar to the make up used by ramp models, in which the image is further draped in additional textures to add an additional layer of detail and believability. These textures are stored in the GPUs memory, where each pixel calculation can access them. This is where memory bandwidth and the quantity of memory plays a vital role which is why we see graphics cards with more memory generally performing better.

Stage 7: Hidden Surfaces

In all 3D scenes, some objects are clearer while some are obscured by others. Its not as simple as writing each pixel to memory. In that case the most recently written triangle would get written first, in which case (for example) a chair placed behind a table would be displayed while the table itself would be hidden behind. This stage involves sorting of triangles from a back to front view. This is where a GPUs *depth buffer* comes in. The depth buffer stores the distance of each pixel from the viewer and before writing a pixel to the screen the GPU does a check to see whether the pixel occupying that position is the closest to the user and if it is, it is sent to the monitor.

Memory

VRAM is present on the PCB of the graphics card and is as important to it as system RAM is to the CPU. Very basically VRAM feeds the GPU with raw data and passes on the results after the GPU has worked on the data. VRAM is also sometimes called ‘frame buffer’. In the past three years or so the importance of VRAM has sharply inclined, mainly because whole visual revolution that is taking place where users want as realistic an experience as possible while playing 3D games. In fact games are the numero uno reason for advancements in graphics cards’ performance over the years. From 32 MB of VRAM on the GeForce 256 DDR, to 768 MB of VRAM on the 8800GTX, the quantity of memory has increased more than 20 fold. This goes in tandem with the improvements to the core of the GPU.

Not only has the quantity of video memory gone up, but the type of memory has changed too. Current graphics cards use DDR3 memory, and although DDR4 memory is waiting in the wings, it hasn’t seen much adoption yet.

The bus width of memory has also gone up, from the 64 bits of yore to 256-bit and even 384-bit bus width (GeForce 8800GTX). All this speed is required too, as today’s games demand more than 50 times the resources than the games released eight years ago. The ultimate goal of manufacturers is to up the memory bandwidth figure of the graphics card. This is done to eliminate bottlenecks which occur in the memory, as 90 percent of the time the core has to wait for much slower memory to supply it with fresh data. Here’s how you calculate the memory bandwidth of your video card (works for system memory too):

An 8800GTX has memory clocked at 1,800 MHz, and this memory is 384 bits wide, the formula for total bandwidth = (clock speed

The importance of VRAM has sharply inclined, mainly because whole visual resolution that is taking place—users want as realistic an experience as possible

Quick Facts

	GeForce 256	GeForce FX 5900	GeForce 7900GTX	GeForce 8800GTX	Radeon X1950XTX	Radeon HD2900XT
Year Of Launch	1999	2004	2006	2006	2006	2007
GPU Name	NV 10	NV 35	G71	G80	R580+	R600
Core Transistor Count	23 million	135 million	278 million	681 million	384 million	700 million
No Of Shader Units	4/0	8/4	24/8	128	48/8	320
Core/Memory speed	120/300 MHz	425/1100 MHz	650/1600 MHz	575/1800 MHz	650 / 2000 MHz	740 / 1650 MHz
Amount Of Memory (MB)	32	256	512	768	512	512
Memory Bus Width	128 bits	256 bits	256 bits	384 bits	256 bits	512 bits

* memory bus width)/8 = (1,800 * 384)/8 = 86,400 megabytes per second or 84.6 GB/s (Gigabytes per second)

Just to compare this figure to the fastest system memory based on PC8500 DDR2 RAM (1066 MHz) we have Bandwidth = (1066*64)/8 = 8528 MB/s or 8.5 GB/s

As you can see current generation cards offer ten times the bandwidth that the fastest memory offers and with DDR4 this bandwidth is set to cross 100 GB/s. This enormous amount of bandwidth is used to feed the data hungry GPU with enough raw data to keep it busy, which is no easy task considering that the GPU has an internal bandwidth of over 180 GB/s, and the latencies are minimal. Very simply, VRAM is a buffer for the GPU. Data that is fed to the GPU is fed via VRAM which holds such data till time the pipelines of the GPU need feeding again. While VRAM holds vertex data, perspective data and lighting information, it can never actually hold texture data which is held by the PCs main memory. The GPU accesses these textures from system memory directly through DMA (Direct Memory Access) and there is no CPU interference. The resultant output of the GPU is also buffered in part of the VRAM which is utilised as a ‘frame buffer’.

RAMDAC

The RAMDAC or Random Access Memory Digital to Analog Converter is a component of a video card that is responsible for converting digital images into analog signals that can be displayed by an analog monitor like a CRT. The RAMDAC may be a separate card on the graphics cards PCB (for example like on the 8800GTX) or could be integrated on the GPU die. The RAMDACs ability to support greater refresh rates depends on its internal data transfer rate, (which depends on its speed, in MHz), and the precision of colour bits used. However the concept of flicker applies only to CRTs and with the advent of the DVI (Digital Video Interface) on most LCD monitors and graphics cards the RAMDACs days are numbered. All LCD and Plasma screens work entirely in the digital domain and do not require a RAMDAC. In case they have an analog input, the signal is simply converted to digital again after the RAMDAC has fed out an analog signal from the original digital signal. This double conversion is lossy to say the least.

Video BIOS

Like a motherboard’s BIOS the VGA BIOS is a chip that contains the firmware (hardwired low-level instructions) that controls the graphics card’s operation and allows the computer’s other components and software to interface with the card. It contains essential information about the graphics card such its core and memory clock speeds, memory timings, volt-ages, and even thermal thresholds (for warnings and shutdowns). Most graphics cards these days come with re-flasheable BIOS’ which may improve performance for affect stability favourably. If the BIOS is flashed incorrectly, the card can be permanently damaged.

Outputs

Connectivity might seem a tiny part of a graphics card, but without standards you wouldn’t see any of that brilliance actually working. The following three connects are common to nearly every graphics card:

HD-515: Commonly known as a D-Sub, or VGA connector. It is an analog connect and has 15 pins and was designed for CRT displays. It has its issues—electrical noise (being an analog signal), image distortion, and sampling errors (due to downscaling).

DVI: A digital standard developed for LCD and Plasma panels. It uses the computers native resolution without any compression of any sort and avoids the demerits of the D-Sub connect.

S-Video: This interface was designed keeping in mind other devices which may need to interface with a display card such as DVD players, large screens, game consoles, video recorders, etc.

Interface

Let’s face it—graphics cards wouldn’t be the speed demons that they are today unless they could interface with PCs using high bandwidth ports. A GPU works in tandem with the CPU, Northbridge (or Memory Controller Hub), and system memory, and its only means of communication happens to be that little slot it’s plugged into. The specifications of the same little slot have changed a lot too over the past few years—so much so that the latest specification is not backwards compatible. From IBMs 16-bit ISA interface with a speed of 8 MHz to the current PCI Express we’ve come a long way. Currently graphics cards utilise the PCI Express interface (also called PCIe or PCI-E). PCIe 1.1 has a maximum bandwidth of 8 GBps, consisting of 32 parallel lanes each capable of a maximum throughput of 250 MBps. PCI 2.0, which has just debuted in the latest ATI Radeon 3xxx series and NVIDIA 8xxx series, offers 16 GBps.

The Shape Of Things To Come

DX 10 wasn’t the big affair it was said to be and we’re not going to say “see we told you so!” Both manufacturers NVIDIA and ATI have realised the follies of developing a graphics card for a standard that has yet to be properly implemented, and the fact that hardly anyone uses Windows Vista as a gaming platform doesn’t help. NVIDIA has just released the first card of their new 9xxx series—yes, the GeForce 9600GT has finally arrived. Based on a 65nm manufacturing process it’s a mid-range offering meant to take over the reins from the capable 8600GT. Rumours abound of the presence of a mighty 9800GTX, but the company themselves are tight lipped about specifications. One thing we know is these cards support DX 10 which is nothing new. ATI has already released a host of products namely the 3xxx series comprising of the high-end Radeon 3870 and the slightly lower Radeon 3850, with the mid-range Radeon 3650. But the specifications of these newcomer’s is hardly revolutionary as the 8800GTX was a couple of years ago. It seems we’ve reached another plateau, and the real meat may arrive after DX 10 matures as a platform and application support picks up. Till then we’re sitting pretty, literally. ■

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